ECE/SIOC 228 Machine Learning for Physical Applications

Lecture 1: Introduction & Course Logistics

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This Course

- Machine learning fundamentals: linear regression, linear classifier, Lasso and Ridge regression, bias and variance tradeoffs in ML
- Classic deep learning models: feedforward/recurrent/convolutional neural networks, optimization and regularization of neural networks, Transformers, etc
- Incorporate Physical and Prior Knowledge into ML/DL: Physics-informed neural networks, neural operators, neural ODEs, Gaussian process, optimization Network (OptNet) and end-to-end learning, etc

Machine Learning for Physical Applications

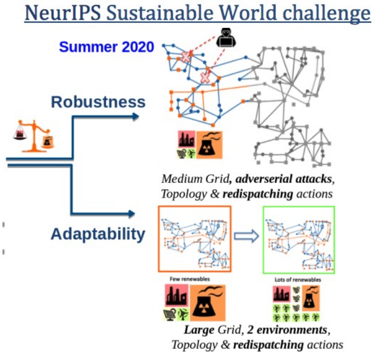


Figure: Machine Learning for power systems (left) and transportation (right).

`http://proceedings.mlr.press/v133/marot21a/marot21a.pdf`

`https://flow-project.github.io/index.html`

Machine Learning for Physical Applications

Figure: Machine Learning for self-driving cars (left) and robotics (right).

https://robohub.org/deep-learning-in-robotics/

https://www.aarp.org/auto/trends-lifestyle/info-2018/self-driving-cars.html

Machine Learning for Physical Applications

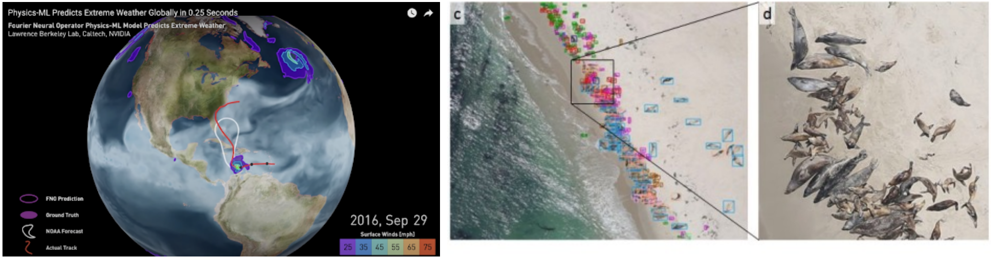


Figure: Machine Learning for extreme weather forecast (left) and marine science (right).

https://www.youtube.com/watch?v=JnGPxZ9glVk&t=11s

https://spo.nmfs.noaa.gov/sites/default/files/TMSPO199_0.pdf

Course Time and Location

- Time: Tuesday, Thursday 12:30 – 1:50pm
- Location: Catalyst Building 0125

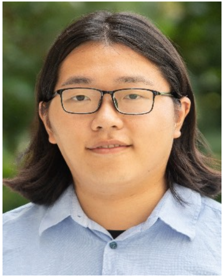
Instruction Team

Main Instructor: Prof. Yuanyuan Shi, yyshi@ucsd.edu
Office Hours: Thursday 2-3pm (Location: Franklin Antonio Hall, Room 2002)

Co-Instructor: Luke Bhan, Email: lbhan@ucsd.edu
Special topics: Neural Operators, Physics-informed Neural Networks, Neural ODE

Lead TA and Course Coordinator: Yuexin Bian, Email: ece228tas@gmail.com

Teaching Assistants



- Benjie Miao, ece228tas@gmail.com, Office Hours: Wednesday 5-6pm (Location: Franklin Antonio Hall, Room 4002)
- Golnaz Salehi, ece228tas@gmail.com, Office Hours: Friday 10-11am (Location: Jacobs Hall Room 5101B)
- Peixing Li, ece228tas@gmail.com, Office Hours: Monday xxx (Location: Jacobs 5101E)

Syllabus

Part 1: Machine Learning / Deep Learning Fundamentals

- Week 1: Course Logistics; supervised learning setup; linear regression
- Week 2: Linear models for classification; Feature selection: ridge regression, Lasso; Bias and Variance Tradeoffs
- Week 3: Neural network basics: deep feedforward neural networks, computational graph and backpropagation; Optimization for training neural networks
- Week 4: Regularization for Deep Learning; Classic Deep Learning architectures: recurrent and convolutional neural networks
- Week 5: Attention & Transformer

Syllabus

Part 2: Specialized Topics: Machine Learning for Physical Applications

- Week 6: Foundation models; Neural ODEs (Taught by Luke)
- Week 7: Neural Operator I; Neural Operator II (Taught by Luke)
- Week 8: Physics Informed Neural Networks (Taught by Luke); Gaussian Process
- Week 9: Optimization Network and End2End Learning; Course Summary
- Week 10: Final project presentation

Why taking this course?

- Learn the foundations of machine learning
- Understand commonly used deep learning models: Feedforward neural networks; recurrent and convolutional neural networks and variants, Transformers
- Special topics on incorporating prior and physics knowledge into deep learning, such as physics-informed neural networks, neural operators, Neural ODEs, Optimization Net, uncertainty quantification to make your DL models more efficient and reliable!
- Hands-on experience in conducting machine learning projects for physical applications.

Prerequisites

- Proficient about coding in Python
- Good foundation in linear algebra, optimization, and probability and statistics.
- Know basics of machine learning, (preferred) have taken some machine learning courses before (e.g., ECE 175, CSE 151, ECE 271, CSE 251 etc)
- If you do not have the requisite background, it is *NOT* recommended for you to take this course. I encourage you to talk to me if you have any doubts.

Several review sessions during TA's office hours next week and will be recorded

- Pytorch
- Linear Algebra and Statistics
- ODEs and PDEs

Grades Breakdown

10% Class Participation (individual)

- Attend lectures in person: occasional end-of-lecture quizzes accessed by scanning a QR code (No score reduction if missing 1 quiz!)

45% Homework (individual)

- 3 written + programming assignments (Roughly Week 2- Week 4; Week 4- Week 6; Week 6- Week 8)
- Late day policy: 2 late days you can use whenever you like (use per day basis); mark the late day usage on the first page of your homework submission. Each late day outside the "2 late days" will lead to a 25% discount on the grade.

45% Final Project (Team)

Course Project

- Team size: up to 4 people (team size 1 is only allowed under exceptional circumstances).
- Two tracks:
  - Reproduce a machine learning + physical / engineering / science application paper and implement some improvement ideas on top of it (the ideas may or may not work)
  - Open-ended project
- Three components:
  - 1-2 pages project proposal with Project Title + Team Member. It will carry NO grade, but is mandatory to submit for early feedback. [Around Week 5]
  - Project oral presentation: $\sim$ 3-min highlight talk [20% of the project score, Week 10]
  - 6-8 pages project final report (NeurIPS template); Include an executable Google Colab or GitHub repo link [80% of the project score, Due on June 10th Midnight]

Project Ideas

- Related to physical/engineering/science applications
- OUT OF SCOPE: For instance, a pure computer vision, natural language processing, or recommender system project without physical/engineering/science applications and motivations is out of scope
- Start early and spend time finding an interesting and well-motivated real-world problem to solve
- Unsure about whether your project idea fits in the class scope? Talk with us!

Previous Years' Project Ideas

Links to previous years' projects:

- 2023: `https://sites.google.com/ucsd.edu/ecesioc-228/project-list-2023?authuser=0`
- 2022: `https://sites.google.com/ucsd.edu/ecesioc-228/project-list-2022`

Course Resources

Canvas

- **Main course website**
- Post all lecture notes, and reading materials
- **We will use Canvas-Discussion for Q&A: Please try to post your questions only on Canvas discussion. It is more efficient and effective than answering emails. It will also benefit your classmates with similar questions!**
- **If you have individual/project-related questions, please email to ece228tas@gmail.com.** Please do not send to my personal email since I may miss your email(s) due to many other emails received every day.
- Help your classmates with their questions and receive bonus participation points (each question you helped answer on Canvas-discussion will add 0.5 bonus point to the course participation category; capped at 10.)

Course Resources

Gradescope

- Post and submit homeworks
- Entry Code: X842RD

Reading Materials

- The second half of this class is advanced topics on Machine Learning for Physical Applications
- Lectures and discussions will be based on research papers
- We have posted all the reading materials on Canvas - Modules - Paper Reading for the 2nd Half of Course. Start reading them at your pace to get prepared

Academic Integrity

- You may discuss problem-solving methods with other students in the class. But, all solutions that are handed in should be written up **individually** and should reflect your own understanding of the subject matter at the time of writing.
- Copying homework done by someone else, copying old homework or the answers online, copying answers from ChatGPT or other generative AI tools without acknowledgment of the resources, and knowingly permitting your work to be copied for the purpose of cheating are all **examples** of cheating.
- **Complete the Acknowledgment of the Code of Conduct before this Friday April 3rd, to start receiving scores in this course.**

Certify Commencement of Academic Activity

- U.S. Department of Education that requires UC San Diego instructors to certify whether students have commenced academic activity at the beginning of each term
- Financial aid awarding requires UC San Diego to document the commencement of academic activity each term. Failure to certify academic activity requires the Financial Aid and Scholarships Office to bill and/or reduce unearned portions of a student's financial aid to ensure UC San Diego remains compliant with federal regulations.
- When to certify: Certification(s) should be submitted no later than Friday, April 10, 2025.
- Where to certify students: submit #FinAid survey under Canvas - Quizzes

Recording Reminder

This class will be recorded by Podcast for educational use by all students in UC San Diego and the recordings are available on Canvas within 1-2 days after each session.

Take Care of Yourself!

- Experiencing stress and mental health issues is a common part of life.
- We all gain from receiving support when facing difficulties.
- Seeking support early can be beneficial.
- You're welcome to utilize my office hours (Thursday 2:00-3:00pm at FAH 2002) for discussions about life, career planning, research ideas, and beyond!

A few final logistic notes

- All homework assignments, office hours of the TAs and instructors start from Week 2
- Finish quizzes and surveys this week:
  - Acknowledgment of the Code of Conduct
  - FinAid survey

Quick Classmates Introduction

`https://docs.google.com/presentation/d/1WQsXA2-lBOOtwdBYyVCv9IIEQc-8fu7qDqgL0I4SMLU/edit?usp=sharing`

Brief Introduction

- Name / advisor or group or major / what do you work on
- Why do choose this course?
- (Optional) Hobbies or anything you'd like to share with the class!

A great way to help me, the teaching team, and your classmates know about you!